

Arithmophysics: Boundary Carry and the Next Signed Estimate

Continuation prompt for the horizon round trip

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Research directive

Continue the existing horizon programme from the proved signed prefix–tail comparison. The immediate task is to estimate the actual signed boundary-and-tail square after the internal prefix history has been removed at a controlled cost. Attempt that estimate directly, and investigate successive interval replacements if they expose additional useful structure.

Use the horizon framework as a working mathematical representation. Move between coefficient space, the continuous field, mode space, and interval boundaries whenever useful. For each passage, give the exact map, its inherited data, its approximation cost, and the return estimate. The cost should determine the next calculation.

An equivalent sufficient condition is a legitimate research target. Establish new estimates from explicitly stated independent inputs. Report the strongest result reached and the precise surviving expression. Keep assessments specific to the method actually analysed.

Read the supplied *Signed Prefix–Tail Comparison: Boundary carry and improved lowest-mode estimates* PDF and its predecessor *Source Truncation and Growing Mode Bounds*. The definitions and essential results below make this directive usable even before the earlier files are available.

1. Fixed source, metric, and closing quantity

For integers $1 \leq J \leq N$, set

$$M(x) = \sum_{n \leq x} \mu(n), \quad c_N(n) = \frac{\mu(n)}{n}, \quad B_N(a, b) = \min(a, b).$$

The original energy, including its terminal, is

$$R(N) = c_N^* B_N c_N = \sum_{k=1}^{N-1} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}.$$

Write $B_N = V_N \Lambda_N V_N^*$, where

$$\theta_{j,N} = \frac{(2j-1)\pi}{2N+1}, \quad v_{j,N}(n) = \frac{2}{\sqrt{2N+1}} \sin(n\theta_{j,N}), \quad \lambda_{j,N} = \frac{1}{4 \sin^2(\theta_{j,N}/2)}.$$

Thus $b_N = V_N^* c_N$, $S_N(J) = \|\Pi_J b_N\|_2^2$, and

$$\frac{N^2}{\pi^2 j^2} \leq \lambda_{j,N} \leq \frac{9N^2}{4j^2}.$$

The objective is a uniform subpower bound for $R(N)$, obtained by controlling the remaining small modes. All constants described as uniform must be independent of N, J .

2. Established boundary replacement: use it as the starting point

For any $1 \leq L \leq N$, let $a_L = M(L)$ and

$$I_L = \sum_{k < L} \frac{M(k)^2}{k(k+1)} = R(L) - \frac{a_L^2}{L}.$$

Let $c_{\leq L}$ and $c_{>L}$ be the actual prefix and tail. Define

$$q = c_{\leq L} - \frac{a_L}{L} e_L, \quad w = c_{>L} + \frac{a_L}{L} e_L, \quad c_N = q + w.$$

The following statements are proved in the supplied PDF:

1. Exact energy accounting.

$$q^* B_N q = I_L, \quad q^* B_N w = 0, \quad R(N) = I_L + w^* B_N w.$$

In the continuous representation

$$\Phi_N(c)(u) = \sum_{n \leq N} \sqrt{n} c_n e^{-|u - \log n|/2},$$

the prefix and its boundary replacement have the same field for $u \geq \log L$. The remainder has zero exterior field.

2. Historical remainder in the modes. Put $r = V_N^* q$ and

$$D_j(k) = \frac{v_{j,N}(k)}{k} - \frac{v_{j,N}(k+1)}{k+1}, \quad \Xi_{N,J}(L) = \sum_{j \leq J} \sum_{k < L} k(k+1) D_j(k)^2.$$

Abel summation gives $r_j = \sum_{k < L} M(k) D_j(k)$, and

$$\|\Pi_J r\|^2 \leq I_L \min\{\Xi_{N,J}(L), \lambda_{J,N}^{-1}\}.$$

In particular,

$$\Xi_{N,J}(L) \leq \frac{4}{2N+1} \frac{L^5 - 1}{45} \sum_{j \leq J} \theta_{j,N}^6, \quad \lambda_{J,N} \Xi_{N,J}(L) \leq \frac{\pi}{45} (L \theta_{J,N})^5.$$

3. Uniform approximation at the required scale. Choose

$$L = L_{N,J} = \left\lfloor (N/J)^{5/6} \right\rfloor, \quad C_\partial = \frac{2\pi^6}{315} = 6.1040583719 \dots$$

Then

$$\|\Pi_J r\|^2 \leq C_\partial \frac{J^2}{N^2}. \tag{B}$$

This follows already from $|\mu(n)| \leq 1$, and holds for arbitrary bounded real charges with their actual inherited total.

4. Improved whole-prefix estimate. For $h = V_N^* c_{\leq L}$,

$$\lambda_{J,N} \|\Pi_J h\|^2 \leq g_{N,J}(L) R(L), \quad g_{N,J}(L) \leq \min \left\{ 1, \frac{4JL}{2N+1} \right\}.$$

The exact optimal factor for arbitrary vectors with the given prefix energy is

$$g_{N,J}(L) = \lambda_{J,N} \lambda_{\max}(G), \quad G_{ij} = \sum_{n=1}^L [v_i(n) - v_i(n-1)][v_j(n) - v_j(n-1)].$$

Here $v_j(0) = 0$. Keep this available when optimising other choices of L .

The boundary charge is retained exactly. Estimate (B) concerns the discarded vector's squared norm; population differences are controlled through the reverse triangle inequality.

3. The next estimate and its complete return

Define the actual retained population

$$W_{N,J} = \frac{4}{2N+1} \sum_{j=1}^J \left[\frac{M(L)}{L} \sin(L\theta_{j,N}) + \sum_{L < n \leq N} \frac{\mu(n)}{n} \sin(n\theta_{j,N}) \right]^2, \quad L = L_{N,J}. \quad (\text{W})$$

The full signed boundary–tail interaction is inside this square. Its relation to the original population is

$$\left| \sqrt{S_N(J)} - \sqrt{W_{N,J}} \right| \leq \sqrt{C_\partial} \frac{J}{N}. \quad (\text{T})$$

For $0 < \eta < 1$, set

$$\ell_N = \log(eN), \quad J_0 = \lceil N/\ell_N^2 \rceil, \quad J_\eta = \min\{J_0, \lceil N^{1-\eta} \rceil\}.$$

The precise remaining target is

$$W_{N,J} \leq B_\eta J^2 N^{-2+\eta} \quad (N \geq 2, \ 1 \leq J < J_\eta), \quad (\text{BC})$$

with B_η uniform. Establishing this for every $0 < \eta < 1$ suffices. A proved partial range, or an independent estimate that combines with the existing range, is also a concrete result.

The predecessor already supplies, for fixed positive constants C, c ,

$$E_N^{\geq J} := \sum_{j \geq J} \lambda_{j,N} |b_N(j)|^2 \leq C \frac{N}{J} \exp \left[-c \sqrt{\log(eN/J)} \right]. \quad (\text{GM})$$

It follows that $E_N^{\geq J_\eta} \leq C(N^\eta + \ell_N^2)$. If (BC) holds, (T) gives $S_N(J) \leq A_\eta J^2 N^{-2+\eta}$, with

$$A_\eta = (\sqrt{B_\eta} + \sqrt{C_\partial})^2.$$

Abel summation on the original fixed vector b_N then yields

$$R(N) \leq \frac{9}{4} A_\eta [1 + 2 \log(3N)] N^\eta + C(N^\eta + \ell_N^2). \quad (\text{Return})$$

Choosing η smaller than the desired exponent gives the subpower energy bound and, through $M(N)^2 \leq NR(N)$, the classical Möbius criterion for RH. The terminal remains included. Since w depends on J , the Abel step must be applied to b_N after using (T).

4. First calculation: keep the boundary and tail together

Estimate (W) directly. Write its boundary square, tail square, and signed mixed term exactly before choosing an inequality. Determine whether the arithmetic saving comes from the tail itself, its interaction with the boundary, or a joint estimate that retains both.

One exact alternative representation, obtained by partial summation, is

$$\frac{M(L)}{L} \sin(L\theta_j) + \sum_{L < n \leq N} \frac{\mu(n)}{n} \sin(n\theta_j) = \frac{M(N)}{N} \sin(N\theta_j) + \sum_{k=L}^{N-1} M(k) \left[\frac{\sin(k\theta_j)}{k} - \frac{\sin((k+1)\theta_j)}{k+1} \right]. \quad (\text{A})$$

Use either representation, or their joint structure, to seek a mean-square or signed-kernel estimate with an independent arithmetic input. Keep the terminal in (A). Taking a maximum of $|M(k)|$ across the interval is an available comparison bound; measure the loss if using it.

Allow L to vary when the historical-error formula certifies an admissible total cost. Optimise the *complete bound*, including the retained square and approximation error. In particular, a zero-return cutoff $M(L) = 0$ is usable when it actually occurs; a uniform argument needs a proved rule for locating or replacing such cutoffs.

A useful result must specify its full N, J, L range and improve an existing bound somewhere. If optimising exponential savings, express all constants against the same stated Mertens or energy input.

5. Second calculation: successive interval replacements

Investigate whether the same mechanism can expose useful relations among boundary increments across several intervals. Start with a partition

$$0 = x_0 < x_1 < \cdots < x_m = N, \quad Q_r = M(x_r) - M(x_{r-1}).$$

On $(a, b] = (x_{r-1}, x_r]$, define

$$c_n^{(r)} = \frac{\mu(n)}{n} \mathbf{1}_{a < n \leq b}, \quad w^{(r)} = \frac{Q_r}{b} e_b, \quad q^{(r)} = c^{(r)} - w^{(r)}.$$

Verify and use the interval analogue of the prefix formula:

$$(V_N^* q^{(r)})_j = \sum_{k=a+1}^{b-1} [M(k) - M(a)] D_j(k), \quad I_r = (q^{(r)})^* B_N q^{(r)} = \sum_{k=a+1}^{b-1} \frac{[M(k) - M(a)]^2}{k(k+1)}. \quad (\text{I})$$

These identities specify the exact inherited data. Empty interval-interior sums vanish.

Carry out the following calculation:

1. Derive the aggregate projection-error bound for $\sum_r q^{(r)}$, retaining cross terms or bounding them explicitly. Disjoint zero-charge remainders are orthogonal in the full B_N metric. Establish the projected estimate separately; also retain interactions with the carried source.
2. Construct and optimise a partition whose accumulated squared error is at most $C_\eta J^2 N^{-2+\eta}$, preferably CJ^2/N^2 , on an explicitly stated range. Dyadic intervals, shorter adaptive intervals, or several retained moments are available constructions. Account for the number of intervals and any growth introduced by combining them.
3. Form the exact retained Gram expression

$$W_{N,J}^{\mathcal{P}} = \sum_{r,s} \frac{Q_r Q_s}{x_r x_s} K_{N,J}(x_r, x_s), \quad K_{N,J}(a, b) = \sum_{j \leq J} v_{j,N}(a) v_{j,N}(b). \quad (\text{P})$$

Seek an inequality for this complete quadratic form using the actual Möbius increments. Identify the arithmetic relation used and propagate it through the kernel.

4. If a recurrence results, give every coefficient, inherited boundary term, error term, and terminal. Compute whether the full recurrence closes at the proposed growth scale.

A partition into singleton intervals has zero approximation error and recovers the original source. Use it as a calibration; quantify what a coarser construction gains. If boundary compression alone supplies only a new representation, identify the additional arithmetic estimate needed and test a specific candidate for it.

6. Use the existing evidence efficiently

Claude's latest report locates substantial slack in the earlier separate prefix and tail bounds. At $(N, J, L) = (16384, 32, 2048)$, it reports weighted prefix population about 0.009 against an allowance 1.50, and weighted tail population about 0.012 against 7.13. The projected interaction was comparatively small in the tested balancing regime. Treat these as diagnostics of that decomposition and range.

The new PDF's following checks have been independently reproduced:

N, J, L	$R(L)/\lambda_J$, old whole-prefix allowance	$I_L \Xi_{N,J}(L)$, historical-remainder allowance
257, 1, 101	5.029876×10^{-5}	1.196549×10^{-7}
1024, 1, 322	3.460066×10^{-6}	2.679424×10^{-9}
4096, 1, 1024	2.162977×10^{-7}	5.528069×10^{-11}
4096, 4, 322	1.060421×10^{-5}	2.057884×10^{-8}

The second column controls the whole prefix; the third controls the history remaining after its boundary charge is retained. At $(4096, 1, 1024)$, the actual prefix–tail inner product is 2.509743×10^{-9} , and its retained boundary contribution is 2.491264×10^{-9} . Keep both signs possible.

For each proposed new inequality, run only the computations needed to check its algebra, constants, and plausible range. Use actual Möbius coefficients, a coherent all-positive control, and random signs on squarefree support with the same normalisation. Record the retained boundary data, exact population, error norm, signed cross terms, and the proposed allowance. This separates geometric approximation from the estimate that uses Möbius arithmetic.

The earlier identities for the log-weighted dilation vector remain available. Their exact collapse locates the arithmetic content; it does not rule out independent estimates for the resulting correlations. Similarly, a failed Mellin-contour estimate or smooth-kernel argument establishes the limitation of that attempted estimate. State any broader obstruction with its precise hypotheses and proof.

7. Required research output

Execute the calculations and write a continuation containing:

1. **The strongest new inequality proved**, with its constants, quantifiers, applicable range, and full proof.
2. **The cost accounting**: what was replaced, which signed boundary data remain, how errors accumulate, and the explicit return to the original population and energy.
3. **The arithmetic input**: the exact step where properties of the actual Möbius source improve on the bounded-charge controls.
4. **Compact reproducible checks**, with code and data for the new claims.
5. **The resulting status of (BC)**: if closed, complete (Return); if partially improved, state the newly controlled range; otherwise exhibit the precise surviving term and the most informative independent inequality attempted.

Preserve the established results in the supplied papers. Add a clearly identified continuation and a short resume note giving the next exact calculation. Proceed with the mathematics now.